

SOP-SG-02 — NFR Definition

Kode	SOP-SG-02
Pemilik	Sogeking (Solution Architect)
Revisi	1.0 · Berlaku 2026-05-29 · Reviu per kuartal
Referensi	ISO/IEC 25010, AWS/Azure Well-Architected (5 pilar), OWASP Top 10, UU PDP, tools/nfr-spec-template.md, tools/well-architected-checklist.md
COBIT	APO03 (Managed Enterprise Architecture), EDM02 (Ensured Benefits Delivery)

1. Tujuan

Memastikan tiap fitur high-stakes punya **NFR (Non-Functional Requirement) yang terukur** — di-define **SEBELUM** build (kontrol **SG-C2**) — silang antara 8 karakteristik **ISO 25010** dan 5 pilar **Well-Architected**, biar gak ada surprise scaling/security/cost di produksi.

2. Ruang Lingkup

- **Berlaku:** fitur high-stakes (sentuh data sensitif, biaya skala, perf user-facing kritikal, atau Type-1), app baru.
- **Tidak berlaku:** fitur low-stakes reversible & internal (NFR cukup pakai default guardrail tim).

3. Definisi & Istilah

- **NFR** — "seberapa bagus" sistem (cepat/aman/skalabel/tahan), bukan "apa fiturnya".
- **High-stakes** — fitur dgn dampak biaya/skala/security/UX besar atau irreversible.
- **RTO / RPO** — Recovery Time Objective (berapa lama boleh down) / Recovery Point Objective (berapa banyak data boleh hilang).
- **Cost envelope** — batas biaya yang disepakati buat fitur pada skala target.

4. Referensi

ISO/IEC 25010 (8 karakteristik mutu). Well-Architected (reliability, security, cost, performance, operational excellence). OWASP Top 10 + UU PDP (security & privacy-by-design). EDM02 (value vs cost). Mandate Tata: **F-1 Backend ALMIGHTY, F-2 Frontend BOOMER-PROOF, no auto-everything-silent** (semua eksplisit + logged).

5. Tanggung Jawab (RACI)

Aktivitas	R	A	C	I
Identifikasi fitur high-stakes	Sogeking	Sogeking	@lelouch	tim
Set target NFR ukur (ISO 25010 × Well-Architected)	Sogeking	Sogeking	@kakashi	tim
Feasibility test/ops NFR	@I, @levi	Sogeking	@I, @levi, @kakashi	tim
NFR yang ngubah scope/biaya	—	@tata	Sogeking	tim

6. Prosedur

6.1 Identifikasi

- 6.1.1 Tentukan apakah fitur **high-stakes** (cek: data sensitif? biaya skala? perf kritikal? Type-1?).
Decision point: low-stakes → pakai guardrail default, keluar SOP. High-stakes → lanjut.

6.2 Set target terukur (ISO 25010 × Well-Architected)

- 6.2.1 Pakai *nfr-spec-template* + *well-architected-checklist*. Set **angka**, bukan kata sifat:
- **Performance efficiency** — latency (mis. p95 < 300ms), throughput.
- **Reliability** — uptime (mis. 99.5%), RTO/RPO.
- **Security** — comply OWASP Top 10 + UU PDP (consent, data residency, privacy-by-design).
- **Cost** — cost envelope per skala target.
- **Maintainability** — coupling / change cost rendah.
- 6.2.2 Contoh *wedding app* — *wishlist kado* + *notif*: p95 notif < 2 dtk, RPO = 0 buat data hadiah (**Data SACRED**, never overwrite), consent UU PDP buat data tamu, cost envelope di tier free.
- 6.2.3 **Decision point:** ada karakteristik ISO 25010 tanpa target ukur? → lengkapi atau tandai *out-of-scope* dgn alasan. Gak boleh ngambang.

6.3 Define-before-build & handoff

- 6.3.1 **Kontrol SG-C2:** NFR spec wajib selesai **sebelum** dev mulai. Exit kalau build sudah jalan tanpa NFR → stop, define dulu (retroaktif = pelanggaran kontrol).
- 6.3.2 **Handoff feasibility** ke @I (target bisa di-test?) & @levi (realistis vs kapasitas ops/infra?). Revisi target kalau infeasible.
- 6.3.3 **Decision point:** NFR ngubah scope/biaya (🟡)? → rekomendasi ke @tata via Pre-Tata Gate. Kalau gak → arsipkan & broadcast.

7. Pengecualian

- **Incident-driven NFR:** kalau gap NFR ketahuan dari incident (input @I/@levi), boleh set target cepat untuk contain, NFR spec formal dilengkapi **< 48 jam**.

8. Rekaman & Retensi

Rekaman	Lokasi	Retensi
NFR spec	arch/nfr-<feature>.md	Permanen
Well-Architected checklist	arch/ / log.md	Per fitur
Feasibility verdict @I/@levi	log.md	Per fitur
Sign-off NFR yg ubah scope	ADR / log.md	Permanen

9. Indikator Kinerja (KPI)

KPI	Rumus	Target
NFR define-before-build	# fitur high-stakes ada NFR sebelum build ÷ total	100%
NFR terukur	# NFR pakai target angka ÷ total NFR	100%
Surprise scaling/security/cost gap di prod	# gap NFR muncul di prod	0
Coverage security (OWASP+UU PDP)	# fitur high-stakes ada review security ÷ total	100%

10. Riwayat Revisi

Rev	Tanggal	Perubahan
1.0	2026-05-29	Terbit pertama